

Spatial modeling improves the calibration of leaf wax hydrogen isotopes to precipitation

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Abstract

Leaf wax hydrogen isotope ratios ($\delta^2\text{H}_{\text{wax}}$) are widely used to infer past precipitation isotopes ($\delta^2\text{H}_{\text{precip}}$) but current global calibrations ignore spatial autocorrelation, inflating apparent relationships. This study compiled 818 sedimentary $\delta^2\text{H}_{\text{wax}}$ measurements and developed hierarchical Bayesian spatial models to quantify spatial structure and its influence on the apparent $\delta^2\text{H}_{\text{wax}}-\delta^2\text{H}_{\text{precip}}$ relationship. Spatial autocorrelation explains 54–68% of total variance and operates at continental scales of 3,400–4,200 km. Accounting for this structure reduced the apparent $\delta^2\text{H}_{\text{wax}}-\delta^2\text{H}_{\text{precip}}$ slope from 0.82 to 0.44–0.61, indicating that much of the observed global correlation reflects broad geographic patterns rather than the precipitation relationship alone.

Spatial effects are dominated by regional intercept shifts, while slopes remain nearly constant worldwide, implying uniform fractionation acting on geographically offset baselines. Vegetation, precipitation, and elevation contribute little once spatial structure is included, indicating that these effects are difficult to resolve given the degree of clustering in the existing dataset. The resulting spatial fields reveal coherent continental-scale patterns. Spatially explicit calibrations improve predictive accuracy by roughly 30% compared to conventional regressions and reduce $\delta^2\text{H}_{\text{precip}}$ reconstruction uncertainty to $\pm 15\%$. In typical sedimentary records, spatial models imply that $\delta^2\text{H}_{\text{wax}}$ changes of $\sim 20\text{--}40\%$ are required to infer real shifts in $\delta^2\text{H}_{\text{precip}}$ at 95% confidence, compared to higher thresholds ($>30\text{--}60\%$) from non-spatial regressions. Model inversions, implemented in the R package ‘leafwax’, provide a framework for geographically informed isotope calibration and more reliable paleoclimate reconstructions.

Keywords: leaf wax, hydrogen isotopes, paleoclimate, spatial autocorrelation

Introduction

Hydrogen isotope ratios preserved in sedimentary leaf waxes ($\delta^2\text{H}_{\text{wax}}$) provide a molecular record of past rainfall and evaporation, making them valuable tools for reconstructing hydroclimate variability from seasonal to millennial timescales (Sachse et al., 2012). These molecules, primarily long-chain *n*-alkanes and *n*-acids produced by terrestrial plants, inherit their isotopic composition from precipitation. The connection to precipitation isotopes ($\delta^2\text{H}_{\text{precip}}$) has enabled researchers to reconstruct past changes in rainfall amount, atmospheric circulation patterns, and moisture sources across diverse environments and time periods (Tierney et al., 2008; Konecky et al., 2016; Daniels et al., 2021; Martins et al., 2022; Zhao et al., 2024).

The relationship between $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$ is complicated by multiple biological, environmental, and sedimentary factors. Different plant functional types (PFTs) exhibit systematic differences in apparent fractionation, with trees, shrubs, and grasses producing distinct isotopic signatures even when drawing from identical water sources (Liu and An, 2019; Konecky et al., 2019a). These differences reflect variations in rooting depth, leaf morphology, and water-use strategies. Additionally, evidence suggests that photosynthetic pathway influences fractionation: C_4 grasses consistently produce more ^2H -enriched waxes than C_3 plants, a difference attributed to variations in leaf water enrichment and transpiration dynamics (Smith and Freeman, 2006). Species-level offsets can span several tens of permil (Liu and An, 2019).

Climate variables beyond precipitation isotopes further modulate the $\delta^2\text{H}_{\text{wax}}$ signal through their effects on evapotranspiration and leaf water enrichment. Temperature influences both the isotopic composition of precipitation through equilibrium fractionation and the degree of evaporative enrichment in leaves (Kahmen et al., 2013; Cernusak et al., 2016). Atmospheric humidity controls transpiration rates and thus the extent of isotopic enrichment during water loss from leaves (Sachse et al., 2012). Soil moisture availability determines which water pools plants access and when they are photosynthetically active, potentially biasing the seasonal signal recorded in waxes (Wu et al., 2016; Goldsmith et al., 2018). While mechanistically important, these climate variables often correlate strongly with $\delta^2\text{H}_{\text{precip}}$ and with each other, creating statistical challenges for attributing their individual effects (Bowen, 2008; Bowen et al., 2019). In sedimentary accumulations, wax signals are integrated across additional scales of complexity. Waxes produced by diverse plants across a catchment undergo mixing during transport to depositional basins, creating a composite signal from an imprecisely defined source area (Galy and Eglinton, 2011). Elevation gradients within catchments introduce isotopic variation through both precipitation lapse rates and potential differences in vegetation composition. Regional climate phenomena such as monsoons, storm tracks, and moisture recycling create spatially coherent patterns in both precipitation isotopes and ecosystem properties that may not be captured by point measurements of environmental variables. Unmeasured environmental and biological parameters contribute additional noise.

These overlapping influences produce considerable scatter in global $\delta^2\text{H}_{\text{wax}}-\delta^2\text{H}_{\text{precip}}$ calibrations (Sachse et al., 2012; McFarlin et al., 2019). The standard approach to calibrating this relationship employs ordinary least squares (OLS) regression (Figure 1), which offers several advantages such as simplicity of implementation, intuitive interpretation, and straightforward uncertainty estimation (McClelland et al., 2021). However, OLS regression assumes that residual variation is random and spatially independent (see details in supplement). This assumption is violated when nearby sites experience similar regional climate patterns, share moisture sources, and often have comparable vegetation assemblages. This spatial autocorrelation has two critical consequences that have been largely overlooked. First, it reduces the effective sample size and inflates uncertainty estimates. Second, it creates spatial confounding where geographic clustering of similar $\delta^2\text{H}_{\text{precip}}$ values inflates the apparent strength of the proxy relationship.

This study demonstrates that spatial confounding fundamentally alters the understanding of leaf wax isotope systematics. Using a global compilation of 818 lake sediment $\delta^2\text{H}_{\text{wax}}$ measurements, we developed hierarchical Bayesian models that explicitly separate spatial structure from environmental relationships. Our analysis reveals that accounting for spatial autocorrelation reduces the identifiable $\delta^2\text{H}_{\text{wax}}-\delta^2\text{H}_{\text{precip}}$ slope by nearly half, from 0.82 to 0.44, indicating that previous calibrations partially reflect geographic covariation as well as the underlying isotopic relationship. Spatial patterns operating at continental scales explain much of the variance in global leaf wax compilations, while vegetation and elevation effects are weakly identifiable once geographic context is considered. These findings have immediate implications for paleoclimate reconstruction: location-specific calibrations can reduce uncertainty by 30%, and models using only paleo-reconstructable variables perform nearly as well as those requiring detailed vegetation data. We provide these models in the R package 'leafwax' to enable spatially-aware reconstructions with proper uncertainty quantification.

2. Methods

2.1 Data Compilation

We compiled hydrogen isotope measurements of sedimentary n -C₂₉ alkanes from 818 globally distributed lake surface sediments (see supplement for detailed methods about data preparation). The n -C₂₉ alkane was selected for its abundance in sediments, demonstrated terrestrial plant origin, and extensive previous calibration efforts, although the approach described here could also be applied to other compounds. Sample locations spanned all inhabited continents (Figure 2), with particularly dense coverage in Asia ($n=391$), moderate coverage in Africa ($n=133$) and the Americas ($n=221$), and limited representation from Europe ($n=32$) and Oceania ($n=41$). Sites ranged from shallow marine sediments at sea level to 4,800 m elevation and encompassed diverse biomes from tropical rainforests to arctic tundra.

For each site, we extracted precipitation hydrogen isotope values ($\delta^2\text{H}_{\text{precip}}$) from the Online Isotopes in Precipitation Calculator (Bowen and Revenaugh, 2003; Bowen, 2018). This raster dataset provides interpolated gridded estimates based on GNIP station data and geographic predictors. We compiled seven additional environmental covariates: elevation from GMTED2010 (Amatulli et al., 2018), vegetation cover fractions (trees, shrubs, grasses) from MODIS land cover (Friedl and Sulla-Menashe, 2019), C₄ vegetation fraction (Luo et al., 2024), and climate variables from TerraClimate (Abatzoglou et al., 2018) including annual precipitation, soil moisture, maximum temperature, and vapor pressure deficit. All vegetation and climate data were averaged over 2001–2019 to reflect long-term patterns relevant to sedimentary integration timescales. Detailed extraction methods and data specifications are provided in Section S2.1.4.

To account for the spatial integration of leaf waxes in sedimentary basins, we implemented multi-scale environmental averaging. For each predictor, we precomputed distance-weighted averages at nine spatial scales (1, 3, 5, 10, 20, 40, 70, 100, 150 km) around each sample site using exponential decay kernels. This integration approach enabled estimation of environmental values at sample locations that fall outside terrestrial raster coverage, such as near-offshore sediment cores and coastal sites, by smoothly averaging across nearby grid cells. Without this spatial averaging, many valuable marine sediment samples would be excluded from analysis due to their positions just beyond the raster boundaries. The multi-scale approach also allows the model to identify the characteristic scale at which environmental signals are integrated into sedimentary leaf waxes, rather than assuming point measurements represent basin-wide conditions.

2.2 Exploratory Data Analysis and Variable Selection

Prior to model development, we conducted exploratory analyses to understand the geographic structure of our data and identify collinear predictors. Examination of residuals from a simple OLS regression of $\delta^2\text{H}_{\text{wax}}$ against $\delta^2\text{H}_{\text{precip}}$ revealed significant spatial autocorrelation (Moran's $I = 0.61$, $p < 0.001$) (Moran, 1950), with systematic regional deviations from the global relationship (see supplement). This spatial structure motivated our modeling approach.

We assessed multicollinearity among environmental predictors using variance inflation factors (VIF; O'Brien, 2007) and correlation analysis. Several climate variables showed strong correlations ($|r| > 0.7$): maximum temperature with vapor pressure deficit ($r = 0.79$), and $\delta^2\text{H}_{\text{precip}}$

with both of these variables ($r = 0.69-0.86$). Based on VIF screening (threshold = 5) (James et al., 2013) and ecological relevance, we retained $\delta^2H_{\text{precip}}$, elevation, C₄ fraction, and plant functional types as primary predictors, while excluding redundant climate variables from our main models (Section S2.2.2). This selection balanced model parsimony with mechanistic interpretability while avoiding numerical instabilities from collinearity.

2.3 Statistical Modeling Framework

We modeled sedimentary δ^2H_{wax} using a hierarchical Bayesian framework (Gelman et al., 2013) that accounts for multiple sources of variation and uncertainty. The fundamental model relates δ^2H_{wax} to environmental predictors through:

$$\delta^2H_{\text{wax},i} \sim \mathcal{N}(\mu_i, \sigma_{\text{total}}^2)$$

(Equation 1)

where μ_i represents the expected value based on environmental conditions and σ_{total}^2 incorporates measurement uncertainty, $\delta^2H_{\text{precip}}$ (OIPC) uncertainty, and residual variation.

The mean function μ_i includes both spatially varying and globally fixed components:

$$\mu_i = \beta_0(s_i) + \beta_{\text{OIPC}}(s_i) \times \delta^2H_{\text{precip},i} + f_{\text{elev}}(\text{elevation}_i) + \beta_{\text{C4}} \times \text{C4}_i + \beta_{\text{precip}} \times \text{precip}_i + \beta_{\text{PFT}} \times \text{PFT}_i + \text{interaction terms}$$

(Equation 2)

The global fixed effects describe the mean influence of:

- **Elevation:** modeled using B-splines with nine interior knots (Wood, 2017) to capture potential nonlinearities without imposing a specific functional form
- **C₄ vegetation fraction:** a linear effect representing the enrichment in C₄-dominated ecosystems
- **Plant functional types:** separate coefficients for tree, shrub, and grass fractions
- **Interactions:** $\delta^2H_{\text{precip}} \times$ vegetation terms testing whether isotopic fractionation varies systematically among plant types.

We explicitly modeled three sources of uncertainty: (1) analytical uncertainty in δ^2H_{wax} measurements; (2) uncertainty in $\delta^2H_{\text{precip}}$ estimates, which were provided by a OIPC uncertainty raster; and (3) residual variation capturing both microscale spatial structure and unexplained variance. This hierarchical formulation propagates all quantified uncertainties through to predictions, yielding realistic confidence intervals for paleoclimate reconstructions. The total modeled variance ($\sigma_{\text{total}}^2 = \sigma_{\text{measurement}}^2 + \sigma_{\text{OIPC}}^2 + \sigma_{\text{residual}}^2$) ensures that predictive intervals reflect both measurement precision and model uncertainty.

The spatially varying coefficients $\beta_0(s_i)$ and $\beta_{\text{OIPC}}(s_i)$ allow slope and intercept of the δ^2H_{wax} response to $\delta^2H_{\text{precip}}$ to vary geographically, as described in the following section.

2.4 Spatial Modeling Framework

To account for geographic variation in the $\delta^2H_{\text{wax}}-\delta^2H_{\text{precip}}$ relationship not captured by environmental predictors, we incorporated spatially varying coefficients into a hierarchical

Bayesian framework. This approach recognizes that nearby sites share similar unmeasured influences that create spatial autocorrelation in the residuals.

The model decomposes the observed $\delta^2\text{H}_{\text{wax}}$ signal into three components:

$$\delta^2\text{H}_{\text{wax}} = \text{Fixed effects} + \text{Spatial field} + \text{Residual variation}$$

(Equation 3)

The fixed effects include the global relationships with $\delta^2\text{H}_{\text{precip}}$, elevation, vegetation, and their interactions. The spatial field captures smooth geographic variation in both the intercept (β_0) and the slope (β_{OIPC}), allowing these parameters to vary continuously across Earth's surface. The residual variation represents local-scale variability and measurement error.

We represented the spatial component using Gaussian processes (GPs; Banerjee et al., 2025), a flexible framework for modeling spatially correlated variation. This approach that has been used in other climate and paleoclimate systems (Tingley and Huybers, 2010; Tierney and Tingley, 2014, 2018) (Supplementary Section 2.4). A GP defines a probability distribution over functions, where the correlation between any two locations depends on their distance. In our application, both the intercept $\beta_0(s)$ and slope $\beta_{\text{OIPC}}(s)$ at any location s vary according to independent Gaussian processes, each with their own spatial correlation structures. This formulation allows the $\delta^2\text{H}_{\text{wax}}-\delta^2\text{H}_{\text{precip}}$ relationship to adapt to regional conditions while maintaining smooth transitions between regions.

The spatial correlation structure is controlled by a length scale parameter (ρ), which determines how quickly correlations decay with distance. Sites separated by distances much smaller than ρ share similar parameter values, while sites farther apart than ρ are effectively independent. Our models estimate ρ from the data, revealing the characteristic scale at which leaf wax isotope patterns vary across the globe. Length scales of 3,400-4,200 km emerged from our analysis, indicating that the $\delta^2\text{H}_{\text{wax}}$ signal integrates processes operating at continental scales. These distances are broadly consistent with the scale of climate systems and atmospheric moisture transport (Gimeno et al., 2012).

For computational tractability with our global dataset, we employed a predictive process approximation (Banerjee et al., 2008; Finley et al., 2009) that projects the full GP onto a reduced set of 125 globally distributed knot locations. This approach maintains the essential spatial structure while reducing computational complexity from $O(n^3)$ to approximately $O(nm^2)$, where $n = 818$ observations and $m = 125$ knots. The knots act as spatial basis functions from which we interpolate the spatial field at any location. We selected knot locations to achieve approximately uniform global coverage, ensuring that regional patterns could be captured even in data-sparse areas.

To prevent overfitting in regions with few observations, we implemented two forms of regularization. First, we applied penalized complexity (PC) priors (Simpson et al., 2017) on the spatial variance hyperparameters, discouraging deviations from a base model with no spatial variation. Second, we implemented spatially-adaptive regularization on individual knot effects (Fuglstad et al., 2019), setting the prior variance to decrease with increasing local data density.

This two-level approach allows the model to determine the overall importance of spatial structure globally while adapting the strength of spatial smoothing regionally, capturing fine-scale variation in data-rich areas while applying conservative smoothing in data-sparse regions (Heaton et al., 2019).

The combination of environmental predictors and spatial fields enables our model to separate process-based relationships from geographic covariation. For instance, if C₄ plants cluster in warm, dry regions that also happen to have enriched $\delta^2\text{H}_{\text{precip}}$ a non-spatial model might attribute too much explanatory power to C₄ vegetation when the true driver is geographic location. By explicitly modeling spatial structure, we can disentangle these confounded relationships and identify the actual contribution of each predictor to $\delta^2\text{H}_{\text{wax}}$ variability (Hodges and Reich, 2010).

2.5 Model Configurations and Comparison

We evaluated 14 model configurations to assess the relative importance of spatial structure, environmental predictors, and their interactions (Table 1). Models ranged from a simple baseline model with only $\delta^2\text{H}_{\text{precip}}$ as a predictor to full specifications incorporating spatial effects, elevation, vegetation, and interaction terms.

Model comparisons used leave-one-out cross-validation information criterion (LOOIC), which estimates out-of-sample predictive accuracy while accounting for posterior uncertainty (Vehtari et al., 2017). Lower LOOIC values indicate better predictive performance. We also computed Bayesian R^2 (Gelman et al., 2019) to assess in-sample fit and root mean squared error (RMSE) for direct performance comparison.

All models were implemented in Stan using Hamiltonian Monte Carlo (Carpenter et al., 2017) with 8 chains of 2,000 to 4,000 iterations (half warmup) depending on model complexity. Convergence was assessed via $\hat{R} < 1.01$ and effective sample sizes > 900 (Vehtari et al., 2021). Models incorporating spatial effects consistently outperformed their non-spatial counterparts, providing decisive evidence for spatial structure (Vehtari et al., 2017). The best-performing model (*full_interact_sp*: LOOIC = 843.2) included spatial effects, elevation, vegetation, and $\delta^2\text{H}_{\text{precip}}$ -Vegetation interactions, though simpler spatial models performed nearly as well.

2.6 Model Validation

To assess potential overfitting and ensure reliable predictions, we conducted multiple validation checks on our spatial models. First, we performed spatial cross-validation (Roberts et al., 2017) by sequentially withholding all observations from each geographic region (Americas, Europe, Africa, Asia, Oceania) and evaluating model performance on the held-out region. The baseline environmental spatial model showed consistent performance across regions, with RMSE values of 10.2‰ (Americas, n=221), 12.0‰ (Europe, n=32), 11.3‰ (Africa, n=133), 18.1‰ (Asia, n=391), and 13.4‰ (Oceania, n=41), compared to an overall RMSE of 14.8‰. The slightly elevated error in Asia likely reflects greater environmental heterogeneity in this region.

To verify that our spatial models captured broad patterns rather than overfitting to local clusters, we examined three diagnostic indicators. First, the estimated GP length scales (3,400-4,250 km) substantially exceeded typical data spacing (mean nearest-neighbor distance = 37 km), ensuring smooth continental-scale patterns rather than localized interpolation. Second, the effective

number of parameters (p_{eff} ; Spiegelhalter et al., 2002) ranged from 43.5 to 56.0 across spatial models, representing only 5.3–6.8% of the 818 observations. This is well below the 10% threshold suggested to avoid overfitting (Piironen and Vehtari, 2017). Third, prediction intervals appropriately widened in data-sparse areas, with mean 95% interval width of 62.0‰ (SD = 4.7‰), a 21% reduction compared to the non-spatial model (78.2‰) while maintaining appropriate coverage.

3. Results

3.1 Model Performance

Evaluation of all 14 model variants (Table 1) revealed a clear hierarchy: spatial models consistently outperformed non-spatial models, while additional environmental and vegetation predictors provided diminishing returns once spatial structure was accounted for.

The best-performing models (*full_sp* and *full_interact_sp*) both achieved identical predictive performance (LOOIC = 843.2) with $R^2 = 0.848$ and RMSE = 15.1‰. This performance represents substantial improvement over the baseline non-spatial model ($R^2 = 0.703$, RMSE = 21.1‰), corresponding to a 28% reduction in prediction error and a 21% increase in explained variance. The addition of a spatial Gaussian process alone (*baseline_sp*) accounted for most of this improvement ($R^2 = 0.831$, RMSE = 15.9‰, $\Delta\text{LOOIC} = 70.7 \pm 22.9$ relative to best model), demonstrating the dominant role of geographic structure in controlling $\delta^2\text{H}_{\text{wax}}$ patterns.

Relatively simple spatial models performed nearly as well as the most complex variants. The baseline environmental spatial model (*baseline_env_sp*), which includes $\delta^2\text{H}_{\text{precip}}$, elevation, and precipitation amount, achieved $R^2 = 0.844$ and RMSE = 15.3‰ ($\Delta\text{LOOIC} = 17.4 \pm 7.5$). This near-equivalence indicates that vegetation predictors (plant functional types, C_4 abundance) and their interactions with $\delta^2\text{H}_{\text{precip}}$ add only minimal predictive power beyond the spatial field and basic environmental variables. Even simpler models using only the spatial field and $\delta^2\text{H}_{\text{precip}}$ (*baseline_sp*: $R^2 = 0.831$, RMSE = 15.9‰) perform nearly as well as models incorporating detailed vegetation information.

Non-spatial models, regardless of predictor complexity, performed substantially worse than even the simplest spatial model. The full non-spatial model incorporating all predictors ($R^2 = 0.746$, RMSE = 19.5‰, $\Delta\text{LOOIC} = 374.6 \pm 44.4$) could not match the *baseline_sp* model that used spatial structure with $\delta^2\text{H}_{\text{precip}}$ alone. This comparison underscores that accounting for geographic autocorrelation is more important than exhaustive characterization of local environmental conditions.

3.2 Spatial Confounding of Variables

A striking result of this study is that the apparent global relationship between $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$ substantially weakens after accounting for spatial autocorrelation. In the baseline non-spatial model (*baseline*), the slope (β_{OIPC}) was 0.818 (95% CI: 0.783–0.855), consistent with ordinary least squares and with recently published global calibrations (Liu and An, 2019; McFarlin et al., 2019). Introducing a Gaussian process spatial field (*baseline_sp*) reduced the slope to 0.608 (95% CI: 0.511–0.712; Table 2; Figure 3), and incorporating environmental covariates further lowered it to 0.444 (95% CI: 0.304–0.576) in the *full_sp* and *full_interact_sp* models. This reduction indicates that nearly half of the apparent correlation in conventional global calibrations reflects spatial confounding rather than direct isotopic coupling.

This spatial confounding arises because both $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$ exhibit broad geographic gradients, producing correlations driven by shared spatial structure rather than the underlying fractionation mechanism. The Gaussian process captured this structure at continental scales. Spatial effects accounted for 54–68% of total variance, far exceeding the contribution of all environmental predictors combined. Nearly all spatial variance (>99%; Table 3) originated from spatially varying intercepts (SD = 14.8–16.6‰) rather than slopes (SD < 0.02), indicating that while baseline $\delta^2\text{H}_{\text{wax}}$ values vary widely across continents (range: –66 to +45‰), the global $\delta^2\text{H}_{\text{wax}}\text{--}\delta^2\text{H}_{\text{precip}}$ slope remains essentially constant.

Together, these results show that conventional global calibrations conflate two distinct components: the true isotopic fractionation process (slope) and regional baseline differences arising from hydrological and ecological integration (spatial intercepts). About half of the apparent $\delta^2\text{H}_{\text{wax}}\text{--}\delta^2\text{H}_{\text{precip}}$ correlation in non-spatial models thus reflects geographic covariation rather than process-level fractionation.

We also tested whether the spatial Gaussian process itself inappropriately absorbed variance from $\delta^2\text{H}_{\text{precip}}$, which would indicate statistical confounding rather than appropriate variance partitioning. To do this, we computed correlations between the posterior mean spatial effects and $\delta^2\text{H}_{\text{precip}}$ values. This correlation was moderate ($r = 0.48\text{--}0.53$, $R^2 = 0.23\text{--}0.28$ across model specifications), indicating that roughly one-quarter of spatial variance is shared with the precipitation isotope gradient, whereas the remaining three-quarters represent independent regional structure.

3.3 Spatial Integration

The spatial Gaussian process exhibited characteristic length scales of 3,400–4,200 km across model variants, indicating that spatial autocorrelation persists over continental distances. The spatial integration scale parameter ($\lambda = 6.8\text{--}8.3$ km) closely matches the OIPC grid resolution (~9 km), suggesting that this parameter reflects optimal handling of gridded climate data rather than revealing true catchment-scale ecological processes. Despite OIPC values remaining correlated beyond 70 km, model performance showed no improvement with larger integration radii, confirming that the estimated λ primarily represents a data resolution constraint rather than an ecologically meaningful integration scale. This pattern holds even in simple OLS regression, where spatial integration consistently improves model fit regardless of modeling approach (Figure S4).

3.4 Environmental Predictors

Despite expectations from vegetation studies, environmental and vegetation predictors showed surprisingly weak effects once spatial structure was accounted for (Table 4). Elevation effects, captured through B-spline basis functions, were minimal across the full elevation range (0–6000 m), although the uncertainty here was very high (Figure S5). In the *baseline_env_sp* model, $\delta^2\text{H}_{\text{wax}}$ values changed by less than 10‰ from sea level to 1000 m elevation, with little additional change at higher elevations. The very wide credible intervals on elevation coefficients indicate that these effects are poorly constrained by the data, with spatial patterns overwhelming any systematic elevation signal.

Mean annual precipitation showed a weak negative relationship with $\delta^2\text{H}_{\text{wax}}$ after accounting for $\delta^2\text{H}_{\text{precip}}$ in non-spatial models, but this effect became negligible in spatial models (coefficient changing from -0.053 to 0.009 in *baseline_env* vs *baseline_env_sp*). This suggests that precipitation amount patterns are already captured by the spatial fields, likely because both precipitation amount and $\delta^2\text{H}_{\text{wax}}$ exhibit strong geographic structure related to continental position and circulation patterns.

Vegetation predictors similarly contributed little explanatory power beyond spatial structure. In non-spatial models, tree cover showed a small negative effect ($\beta = -0.003$ to -0.005 , corresponding to $\sim 3\text{-}5\%$ change per 10% increase in cover), and grass cover showed a comparably small effect ($\beta = -0.002$). However, upon adding spatial structure, only grass cover maintained marginal significance ($\beta = -0.002$, 95% CI: -0.003 to -0.001), while tree and shrub effects became non-significant. C_4 plant abundance showed no consistent effect across model configurations, with coefficient estimates ranging from -0.069 to $+0.111$ and wide credible intervals overlapping zero in most models.

Plant functional type interactions with $\delta^2\text{H}_{\text{precip}}$ were tested in the *full_interact_sp* model to assess whether vegetation modulates the isotopic relationship. All interaction terms showed negligible effects: $\text{OIPC} \times \text{C}_4$ ($\beta = -0.015$, 95% CI: -0.124 to 0.094), $\text{OIPC} \times \text{tree}$ ($\beta = 0.040$, 95% CI: -0.019 to 0.099), $\text{OIPC} \times \text{shrub}$ ($\beta = 0.066$, 95% CI: -0.031 to 0.163), and $\text{OIPC} \times \text{grass}$ ($\beta = 0.070$, 95% CI: -0.005 to 0.145). The lack of significant interactions indicates that the $\delta^2\text{H}_{\text{wax}}\text{-}\delta^2\text{H}_{\text{precip}}$ relationship remains consistent across landscapes dominated by different vegetation types, contrary to expectations from plant physiological studies suggesting strong vegetation-specific fractionation factors (Smith and Freeman, 2006; McNerney et al., 2011; Sachse et al., 2012; Kahmen et al., 2013).

4. Discussion

4.1 Spatial confounding limits global calibrations

The most important result of this study is that spatial confounding accounts for approximately half of the apparent global correlation between $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$. When spatial autocorrelation is properly incorporated through Gaussian process models, the slope of the $\delta^2\text{H}_{\text{wax}}\text{-}\delta^2\text{H}_{\text{precip}}$ relationship decreases from 0.82 in conventional analyses to 0.44 in our best spatial models. This result does not indicate that previous calibrations were incorrect, but rather that they conflated two distinct processes: the direct isotopic relationship between $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$, and geographic covariation arising from the spatial structure of both variables.

Spatial confounding occurs when a spatially structured predictor correlates with other regional processes that also influence the response variable. In conventional non-spatial calibrations of $\delta^2\text{H}_{\text{wax}}$ versus $\delta^2\text{H}_{\text{precip}}$, the regression coefficient conflates the direct isotopic relationship between $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$ with geographic covariation arising because both variables exhibit spatial structure. Regions with similar $\delta^2\text{H}_{\text{precip}}$ values cluster spatially, and these same regions also share regional characteristics (moisture sources, evapotranspiration regimes, vegetation composition) that influence the $\delta^2\text{H}_{\text{wax}}\text{-}\delta^2\text{H}_{\text{precip}}$ relationship. Non-spatial models cannot separate these effects, so the precipitation coefficient absorbs variance from regional patterns, inflating the apparent strength of the relationship. Our recovered apparent slope of ~ 0.44 is the component of this relationship that remains after accounting for spatial structure.

Our spatial models appropriately partition this variance by explicitly modeling regional structure through the Gaussian process. The $\delta^2\text{H}_{\text{precip}}$ coefficient now captures the direct process-based relationship, while the spatial field accounts for continental-scale baseline differences. However, introducing spatial structure creates a different concern: that the spatial GP might itself be confounded with $\delta^2\text{H}_{\text{precip}}$, inappropriately absorbing variance that should be attributed to the precipitation effect. This would make our spatial models unreliable, i.e. the slope reduction could reflect statistical artifact rather than genuine variance partitioning.

Correlations between the estimated spatial effects and $\delta^2\text{H}_{\text{precip}}$ show that the spatial term is not simply capturing the precipitation signal. These correlations are moderate ($r = 0.48\text{--}0.53$), indicating that roughly 72–77% of the spatial variance is independent of the precipitation isotope gradient (Hodges and Reich, 2010). This confirms that the slope reduction from 0.82 in the non-spatial model to 0.44–0.61 in the spatial models reflects proper variance partitioning rather than confounding. In the non-spatial models, $\delta^2\text{H}_{\text{wax}}\text{--}\delta^2\text{H}_{\text{precip}}$ relationships are inflated because regional offsets are absorbed into the slope. The spatial models correct this by representing those regional differences through varying intercepts. Consequently, non-spatial calibrations overestimate the strength of the $\delta^2\text{H}_{\text{wax}}\text{--}\delta^2\text{H}_{\text{precip}}$ relationship by approximately a factor of two.

Future calibration efforts should account for spatial structure explicitly, either through spatial regression models or by developing regionally specific calibrations that minimize geographic confounding within more homogeneous climate zones. The substantial improvement in predictive performance (R^2 increasing from 0.70 to 0.85, RMSE decreasing from 21‰ to 15‰) demonstrates that spatially aware approaches provide more accurate and precise predictions. However, this improvement comes with important caveats about spatial stationarity that are addressed in section 4.5.

4.2 Signals integrate over continental scales

The Gaussian process length scales estimated by our models (3,400–4,200 km) represent the characteristic distance over which spatial autocorrelation decays: essentially, how smoothly the spatial field interpolates between observations (Figure 4). The similarity of these estimates across model specifications indicates this is a robust statistical feature of our dataset. These length scales should not be directly interpreted as physical process scales. They describe the smoothness of the residual spatial field after accounting for measured covariates, not the spatial extent of any particular climatic process.

Although the length scales are purely statistical parameters, the spatial patterns they capture reveal clear regional structure. Western North and South America are systematically depleted relative to the global mean, whereas Europe and eastern Asia are enriched, with considerable within-region variability. These residual baseline differences remain after accounting for $\delta^2\text{H}_{\text{precip}}$, elevation, vegetation, and precipitation amount. This pattern may arise from a combination of geographic factors, including differences in moisture source trajectories, evapotranspiration regimes, the seasonal integration of precipitation isotopes by plants, soil-water dynamics, and other unmeasured regionally varying influences.

The dominance of intercept variation over slope variation provides crucial insight into what these spatial patterns represent. Different regions share essentially the same sensitivity of $\delta^2\text{H}_{\text{wax}}$ to $\delta^2\text{H}_{\text{precip}}$ but exhibit large, persistent baseline offsets. The minimal slope heterogeneity argues against regionally varying fractionation factors; instead, regional hydrological and ecological processes create baseline differences in $\delta^2\text{H}_{\text{wax}}$ values that are consistent within regions but differ between them.

What drives these regional baselines is not fully understood. Possible factors include differences in moisture recycling, growing-season timing, how vegetation samples seasonal precipitation, soil-water residence times that smooth precipitation signals, and vegetation effects too coarse to capture with global datasets. The large length scales suggest these patterns vary smoothly across space, consistent with gradual transitions in climate regimes rather than discrete regional processes.

The spatial integration scale for precipitation isotopes ($\lambda = 7\text{-}8\text{ km}$) operates at a fundamentally different scale than the Gaussian process length scales. This integration scale corresponds closely to the OIPC grid resolution (0.083°) and likely reflects methodological optimization rather than geochemical processes. Integrating over several pixels rather than using point estimates reduces sensitivity to pixel assignment uncertainty when sample locations fall near grid boundaries, improving model fit even in non-spatial models (R^2 from 0.653 to 0.703). The GP length scales, in contrast, describe the spatial structure of residual variation after accounting for all measured predictors—they capture real patterns we cannot explain with our covariates, though what specific processes create that structure remains uncertain.

For proxy interpretation, the key insight is that location matters. The same $\delta^2\text{H}_{\text{wax}}$ value translates to different $\delta^2\text{H}_{\text{precip}}$ depending on geographic context, with offsets of $\pm 30\text{-}60\text{‰}$ across the globe. Our models capture this spatial structure empirically through the Gaussian process, enabling better predictions even without complete mechanistic understanding of what drives regional baselines. This has practical implications: $\delta^2\text{H}_{\text{wax}}$ records are most reliably interpreted when the site's modern spatial context is known, allowing appropriate baseline adjustments. However, this approach depends on spatial stationarity, the assumption that current patterns remain constant over time. If that assumption fails, the true prediction uncertainty would be substantially greater than our model estimates.

4.3 The vegetation paradox

The weak global influence of vegetation on $\delta^2\text{H}_{\text{wax}}$ contrasts with previous results showing that plant functional type exerts a major control on hydrogen isotope fractionation. Previous studies demonstrate clear differences: C_4 plants produce waxes enriched by more than 20‰ relative to C_3 plants under identical conditions (Smith and Freeman, 2006; Sachse et al., 2012; Shanahan et al., 2013), and trees, shrubs, and grasses show distinct fractionation factors reflecting differences in rooting depth, leaf morphology, and transpiration dynamics (Hou et al., 2007; Polissar and Freeman, 2010; Kahmen et al., 2013; Konecky et al., 2019a). Yet in our global dataset, these vegetation effects are either non-significant (C_4 abundance, tree and shrub cover) or very small (grass cover showing -0.002‰ per percent cover change). How can we reconcile this apparent contradiction?

The answer lies in understanding spatial confounding and scale-dependent processes. Vegetation types are not randomly distributed across the globe but cluster geographically in patterns that correlate strongly with climate. C₄ grasses dominate in warm, seasonal tropical and subtropical regions where $\delta^2\text{H}_{\text{precip}}$, evapotranspiration, and soil moisture dynamics also differ systematically from temperate grasslands or boreal forests. When we introduce spatial structure into our models, the Gaussian process and the C₄ covariate compete to explain the same geographic patterns. The spatial model wins this competition not because vegetation effects don't exist, but because at global scales with the current sampling density and covariate resolution, we cannot reliably separate the direct physiological effects of C₄ metabolism from the regional climate characteristics of areas where C₄ plants thrive.

This spatial confounding is evident in the behavior of the C₄ coefficient, which shifts from -0.069 in non-spatial models to $+0.049$ in spatial models, with wide credible intervals overlapping zero in most cases. Such instability suggests the coefficient is not capturing a consistent biological signal but is instead fluctuating as it competes with the spatial term for shared variance. The moderate correlation between spatial effects and $\delta^2\text{H}_{\text{precip}}$ ($r \approx 0.5$) leaves enough independent spatial variation for vegetation-related signals to be partially absorbed by the spatial field. A mismatch of spatial scales contributes to this outcome. The spatial integration of precipitation isotopes ($\lambda \approx 7\text{--}8$ km) smooths over fine-scale vegetation heterogeneity, as sedimentary catchments average signals across local plant communities. In addition, global vegetation datasets are too coarse (C₄ at 56 km; PFT at 9 km) to resolve small-scale variation that might drive detectable isotopic effects. Sedimentary mixing further compounds this by integrating wax inputs across both space and time, reducing the expression of vegetation heterogeneity in the sedimentary record.

The implications for paleoclimate applications are both encouraging and cautionary. On one hand, reconstructions need not account for detailed vegetation changes if the spatial context of a site remains constant. The universal slope means that the fundamental $\delta^2\text{H}_{\text{wax}}\text{--}\delta^2\text{H}_{\text{precip}}$ relationship holds regardless of whether vegetation shifts from C₃ to C₄ or from forest to grassland. This simplifies inversions and reduces uncertainty propagation. On the other hand, this result reflects an analytical constraint of global-scale spatial modeling rather than definitive evidence that vegetation effects are truly negligible. Local-scale studies with dense spatial sampling may resolve vegetation signals that global models cannot detect. The minimal grass effect we do detect (-0.002‰ per percent) suggests vegetation influences exist but are overwhelmed by regional baseline differences in globally integrated datasets.

These results indicate that for most applications, reconstructing past C₃/C₄ vegetation fractions is unnecessary when using spatial calibrations. The vegetation effects are too small relative to total prediction uncertainty (RMSE $\sim 15\text{‰}$; 95% prediction intervals averaging 62‰) to justify the added complexity of paleovegetation reconstruction.

An exception arises when spatial stationarity cannot be assumed. For example, at sites that underwent major biome transitions such as a C₃ forest to C₄ grassland shift, independent evidence for vegetation change may be required to evaluate whether local isotopic baselines remained stable. For paleoclimate reconstructions where vegetation type has remained broadly

similar, spatial models offer the best performance without the need for detailed paleoecological reconstructions.

4.4 Practical Applications

4.4.1 Uncertainty Reduction and Spatial Context

Spatial models reduce prediction uncertainty by roughly 30% compared to conventional global regressions (RMSE $\approx 15\%$ vs 21%). This improvement arises from accounting for location-specific baseline offsets that, if ignored, inflate prediction error. For a paleorecord, incorporating geographic context allows the model to apply an appropriate regional intercept adjustment, tightening uncertainty bounds on $\delta^2\text{H}_{\text{precip}}$ reconstructions.

4.4.2 Model selection for paleoclimate reconstruction

Simplified spatial models designed for paleoclimate use (*elevation_only_sp* and *elevation_c4_sp*) perform nearly as well as the full model ($\Delta\text{LOOIC} = 17\text{--}18$ vs 0 for *full_interact_sp*) while relying only on variables obtainable from sedimentary archives. Elevation can be constrained through basin analysis or geomorphology, and C_4 abundance inferred from $\delta^{13}\text{C}_{\text{wax}}$ measurements. This near-equivalent performance shows that reliable reconstructions can be achieved without detailed precipitation or vegetation inputs, which are rarely available independently.

4.4.3 Detecting climate change signals

This reduction in uncertainty translates directly into improved ability to detect climate shifts in sedimentary records. The threshold for confidently detecting a change depends on both calibration uncertainty and the temporal autocorrelation structure of the archive. Within a single record where spatial stationarity holds, spatial effects cancel when comparing time intervals, leaving analytical uncertainty and residual variance as the dominant error terms.

For an autocorrelated series, the variance of the difference between two samples is $\text{Var}(X_1 - X_2) = 2\sigma^2(1 - \rho)$ (Box et al., 2013). With a residual standard deviation of $\sim 14\%$ for spatial models and analytical uncertainty of $\sim 3\%$, the combined uncertainty is $\sqrt{(14^2 + 3^2)} \approx 14.3\%$. For independent samples ($\rho = 0$), the standard deviation of differences becomes $\sqrt{2} \times 14.3 \approx 20.2\%$. To be 95% confident that an observed change in $\delta^2\text{H}_{\text{wax}}$ reflects a real change rather than measurement and model uncertainty, the change must exceed approximately $1.96 \times 20.2 \approx 40\%$. However, sedimentary records typically exhibit positive temporal autocorrelation (Huybers and Wunsch, 2004) from sedimentological processes including catchment storage and soil residence time (Galy and Eglinton, 2011; Tao et al., 2015), bioturbation (Boudreau, 1994) and sediment time-averaging from slow accumulation rates (Sadler, 1981).

With moderate autocorrelation ($\rho = 0.5$), observed shifts in $\delta^2\text{H}_{\text{wax}}$ must exceed $\sim 28\%$ to confidently infer (at 95% confidence) that $\delta^2\text{H}_{\text{precip}}$ changed at all, while strong autocorrelation ($\rho = 0.8$) reduces this detection threshold to $\sim 18\%$ (Figure 5). For a single sedimentary record, changes in $\delta^2\text{H}_{\text{wax}}$ smaller than these thresholds cannot be statistically distinguished from calibration uncertainty alone.

4.4.4 Implementation and uncertainty propagation

The accompanying R package **leafwax** implements Bayesian inversion to reconstruct $\delta^2\text{H}_{\text{precip}}$ from $\delta^2\text{H}_{\text{wax}}$ measurements, providing full posterior distributions that propagate all sources of uncertainty including analytical measurement error, spatial uncertainty in site-specific effects, residual variance after accounting for predictors, and uncertainty in global regression parameters. These posterior distributions enable probabilistic inference rather than binary hypothesis testing, allowing users to calculate the probability that $\delta^2\text{H}_{\text{precip}}$ exceeds any threshold of interest or that changes between time periods are meaningful.

The **leafwax** package can be used with Bayesian time point packages like **bcp** (Erdman and Emerson, 2007) or **Rbeast** (Zhao et al., 2019), which provide sophisticated frameworks for estimating temporal autocorrelation and calculating posterior probability of change of a given magnitude. While these packages can help estimate temporal autocorrelation, the fundamental detection thresholds shown in Figure 5 represent inherent statistical limits imposed by calibration uncertainty.

4.4.5 Limits of spatial stationarity

All applications described in this section rely on the assumption of spatial stationarity, i.e. that modern spatial covariance patterns remain valid through the reconstruction interval. This assumption is most defensible for late-Holocene and glacial–interglacial transitions, when circulation patterns and continental geometry were broadly similar to today (Liu et al., 2009; Clark et al., 2009). For older intervals or abrupt reorganizations of atmospheric circulation such as Heinrich events (Bond et al., 1993), Dansgaard-Oeschger cycles (Dansgaard et al., 1993) or monsoon collapses (Wang et al., 2005), spatial structure may have differed substantially from modern patterns, potentially biasing reconstructions. Deep-time applications involving major continental rearrangements, altered topography, or fundamentally different climate states present additional challenges to stationarity assumptions. Identifying when and why spatial patterns change remains a key target for future work.

4.5 Limitations and future directions

Several limitations constrain the interpretation and application of our results. Sampling gaps limit the ability to characterize spatial patterns in underrepresented climate zones such as the tropics and are likely contribute to elevated prediction errors in Asia.

Temporal validation remains critical. Testing whether calibration relationships hold through time would require archives with independent $\delta^2\text{H}_{\text{precip}}$ constraints (e.g., speleothem fluid inclusions, ice cores) that can be compared with coeval leaf wax records to assess whether slopes, intercepts, or spatial structure vary systematically through major climate transitions. While such multi-proxy comparisons remain rare, they represent a critical direction for validating stationarity assumptions.

The coarse resolution of global environmental datasets introduces unquantified uncertainty. C_4 vegetation (56 km resolution), plant functional types (9 km), and interpolated precipitation isotopes may miss ecologically relevant heterogeneity. Higher-resolution datasets, particularly in topographically complex regions, could improve calibration precision and help separate biological from climatic signals. Recent advances in land cover classification using Sentinel-2 imagery and deep learning have achieved 10m resolution (Karra et al., 2021), substantially finer

than the ~9 km MODIS-derived products used here. These products currently lack the multi-decadal temporal coverage needed to characterize long-term vegetation patterns but will be important in the future. Future calibrations incorporating both high spatial and temporal resolution vegetation data, combined with temporally dynamic precipitation isotope models (Yoshimura et al., 2008) may help constrain seasonal integration effects and better resolve vegetation controls in topographically complex regions.

5. Conclusions

Spatial autocorrelation fundamentally shapes global leaf wax hydrogen isotope patterns, accounting for 54-68% of variance in $\delta^2\text{H}_{\text{wax}}$ and operating at continental scales ($\lambda = 3,400\text{-}4,200$ km). This spatial structure profoundly affects our interpretation of the $\delta^2\text{H}_{\text{wax}}\text{-}\delta^2\text{H}_{\text{precip}}$ relationship. When spatial effects are properly incorporated through Gaussian process models, the apparent global slope decreases from 0.82 to 0.44-0.61, revealing that up to half of the observed correlation in conventional calibrations reflects geographic covariation rather than direct isotopic processes. This spatial confounding arises because geographic variables such as vegetation distributions, climate regimes, and continental positioning cluster spatially and correlate with both $\delta^2\text{H}_{\text{wax}}$ and $\delta^2\text{H}_{\text{precip}}$, inflating apparent process relationships in non-spatial analyses.

Geographic location explains substantially more variance than vegetation or elevation. Despite evidence for large fractionation differences among plant functional types, vegetation effects are minimal or inconsistent in our global spatial models. This reflects the dominance of regional baseline offsets over local ecological controls, although this conclusion may be partly driven by the coarse resolution of global vegetation rasters. The spatial patterns reveal universal slopes but regionally varying intercepts: the fundamental $\delta^2\text{H}_{\text{wax}}\text{-}\delta^2\text{H}_{\text{precip}}$ relationship remains consistent globally (slope coefficient of variation ~2%), while baseline $\delta^2\text{H}_{\text{wax}}$ values vary dramatically across continents (intercept SD = 14-17‰). This implies that average hydrogen isotope fractionation during leaf wax synthesis is essentially constant worldwide, while regional hydrological processes create large but predictable spatial offsets.

In paleoclimate applications, changes in $\delta^2\text{H}_{\text{wax}}$ are often interpreted to reflect corresponding changes in $\delta^2\text{H}_{\text{precip}}$. This interpretation requires assumptions about calibration error and spatial structure that are difficult to quantify without a statistical framework. The models developed here provide this framework. For paleoclimate applications, spatial models reduce reconstruction uncertainty by approximately 30% (reducing RMSE from ~21‰ to ~15‰) compared to conventional regressions by accounting for persistent location-specific biases. This improvement translates to substantially enhanced ability to detect genuine climatic shifts in sedimentary records, with detection thresholds dropping from ~40‰ to ~20‰ for moderately autocorrelated time series. The **leafwax** R package implements these spatial calibrations and provides full posterior distributions for paleoclimate reconstruction with rigorous uncertainty propagation, enabling the paleoclimate community to apply these methods to their own records.

Acknowledgments

The author acknowledges support from a block grant from the Taylor Geospatial Institute. Washington University provided support from the Geospatial Research Initiative and from the Department of Earth, Environmental, and Planetary Sciences.

Data Availability

All data and code are available through GitHub. The spatial model is available at: <https://github.com/bradleylab/leafwax-spatial> and the leafwax R package is available at: <https://github.com/bradleylab/leafwax>

Appendix A: Supplementary Materials

The supplementary material provides additional methodological details for the Bayesian spatial models used in this study. It includes full descriptions of the model design, data processing steps, and prior specifications.

Figure Captions

Figure 1. Global relationship between precipitation and leaf wax hydrogen isotopes.

Ordinary least squares regression of $\delta^2\text{H}_{\text{wax}}$ against $\delta^2\text{H}_{\text{precip}}$ using point values ($n = 818$ sediment samples). The linear relationship ($y = -126.9 + 0.802x$) explains 65.3% of variance ($R^2 = 0.653$, $\text{RMSE} = 22.8\text{‰}$). Shaded region represents 95% confidence interval.

Figure 2. Global distribution of sediment samples and environmental variables. (A) $\delta^2\text{H}_{\text{precip}}$ (OIPC), (B) Annual precipitation, (C) Maximum temperature, (D) Vapor pressure deficit, (E) Soil moisture, (F) Elevation, (G) C_4 vegetation fraction, and (H) Plant functional type. Red points show the 818 sediment sample locations used in this study. Environmental data are extracted at sediment locations from global gridded datasets (see Methods). Red points indicate calibration sample locations.

Figure 3. Spatial confounding inflates the apparent $\delta^2\text{H}_{\text{wax}}$ - $\delta^2\text{H}_{\text{precip}}$ relationship. (A) Non-spatial regression of $\delta^2\text{H}_{\text{wax}}$ against $\delta^2\text{H}_{\text{precip}}$ (slope = 0.82, 95% CI: 0.79–0.86) showing an artificially steep relationship driven by continental-scale geographic covariation. Points are colored by continent, highlighting regional clustering. (B) Spatially adjusted regression after removing Gaussian-process spatial effects ($\delta^2\text{H}_{\text{wax, adj}} = \delta^2\text{H}_{\text{wax, obs}} - \alpha_{\text{spatial}}(s)$, where $\alpha_{\text{spatial}}(s) = \beta_0(s) + \beta_{\text{OIPC}}(s) \times \delta^2\text{H}_{\text{precip}}$). The spatial model yields a flatter relationship (slope = 0.61, 95% CI: 0.52–0.72), representing the underlying isotopic fractionation after accounting for regional baseline offsets.

Figure 4. Geographic patterns of spatially varying model parameters. Posterior mean estimates from the *full_sp* model showing (A) spatially varying intercepts (β_0) representing location-specific baseline $\delta^2\text{H}_{\text{wax}}$ values, and (B) spatially varying slopes (β_{OIPC}) representing location-specific sensitivity of $\delta^2\text{H}_{\text{wax}}$ to $\delta^2\text{H}_{\text{precip}}$. Black crosses indicate knot locations; colors represent interpolated values between knots.

Figure 5. Detection thresholds for identifying shifts in leaf wax $\delta^2\text{H}$ values. Minimum detectable difference (‰) versus confidence level for varying spatial autocorrelation (ρ). Solid lines show the spatial model; dotted lines show non-spatial model predictions. At 95% confidence, detection thresholds range from ~17‰ (spatial model, $\rho = 0.9$) to ~77‰ (non-spatial model, $\rho = 0$). Spatial models consistently outperform non-spatial models across all

autocorrelation levels, with the greatest improvement at low autocorrelation. Higher spatial autocorrelation reduces detection thresholds in both model types.

Table titles

Table 1. Model performance metrics for all candidate models.

Table 2. Global parameter estimates for all models.

Table 3. Variance decomposition for spatial models.

Table 4. Environmental covariate coefficients.

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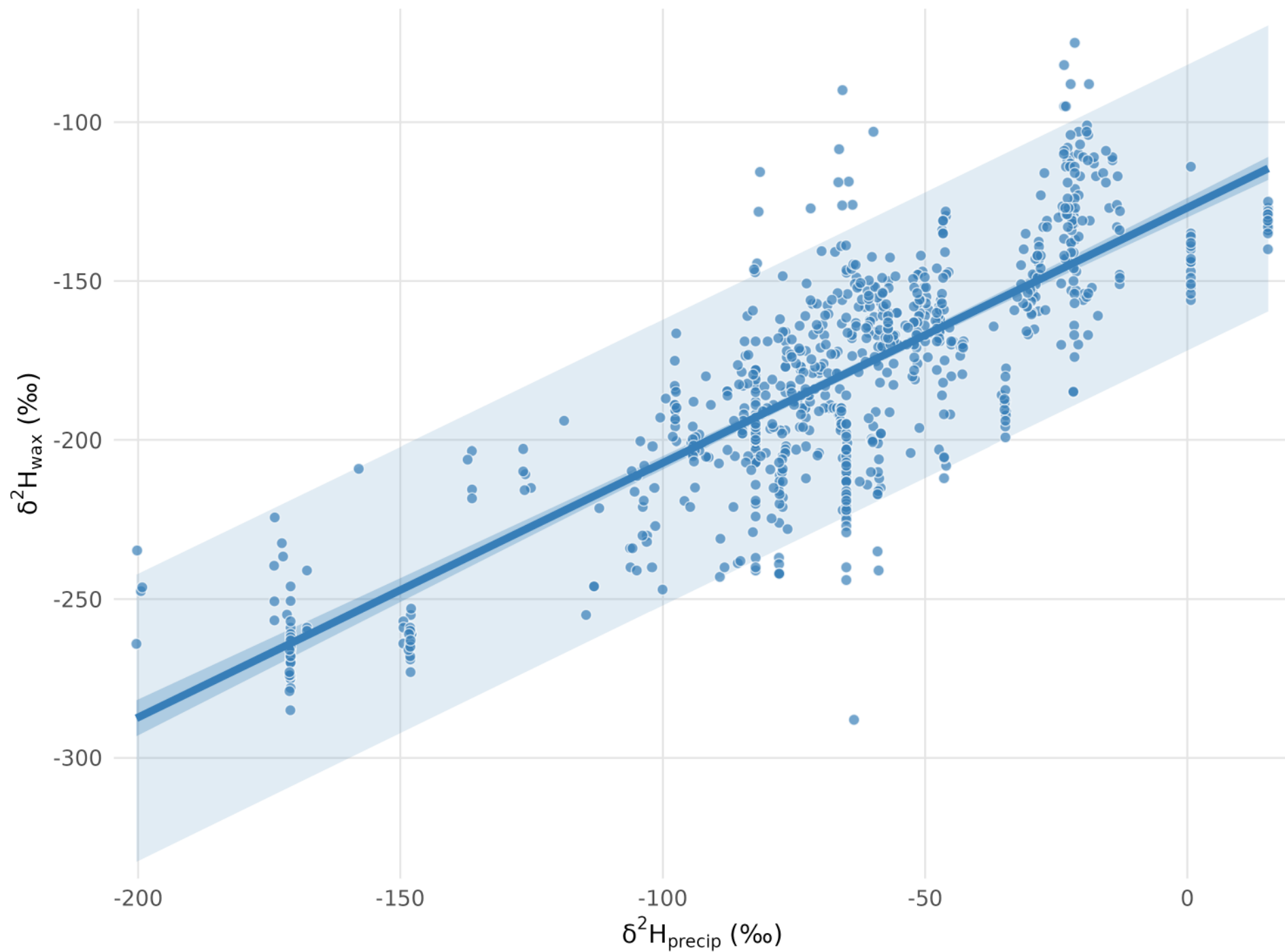
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OLS regression (point values)

$y = -126.9 + 0.802x$, $R^2 = 0.653$, $RMSE = 22.8\text{‰}$, $n = 818$



A world map showing annual precipitation in millimeters (mm). The map uses a color scale from light yellow (100 mm) to dark green (3000 mm). The map includes latitude and longitude markings. A legend at the bottom indicates the color scale for annual precipitation (mm).

Annual precip (mm)

100 300 1000 3000

World map showing VPD (hPa) distribution. The map uses a color scale from 0 (lightest) to 4 (darkest) to represent VPD values. Red dots are scattered across the map, indicating specific data points. The map includes latitude and longitude markings and a color bar at the bottom.

A world map showing elevation and sampling locations. The map uses a color scale from green (0m) to brown (5000m) to represent elevation. Red dots indicate sampling locations across various continents, including North America, South America, Europe, Africa, Asia, and Australia. The map includes latitude and longitude markings and a legend for elevation in meters.

A world map showing the distribution of vegetation types. The map is color-coded: yellow for Grass, light green for Shrub, and dark green for Tree. Red dots are scattered across the map, indicating specific locations. The map includes latitude and longitude markings. A legend in the bottom left corner identifies the colors: Grass (yellow), Shrub (light green), and Tree (dark green).

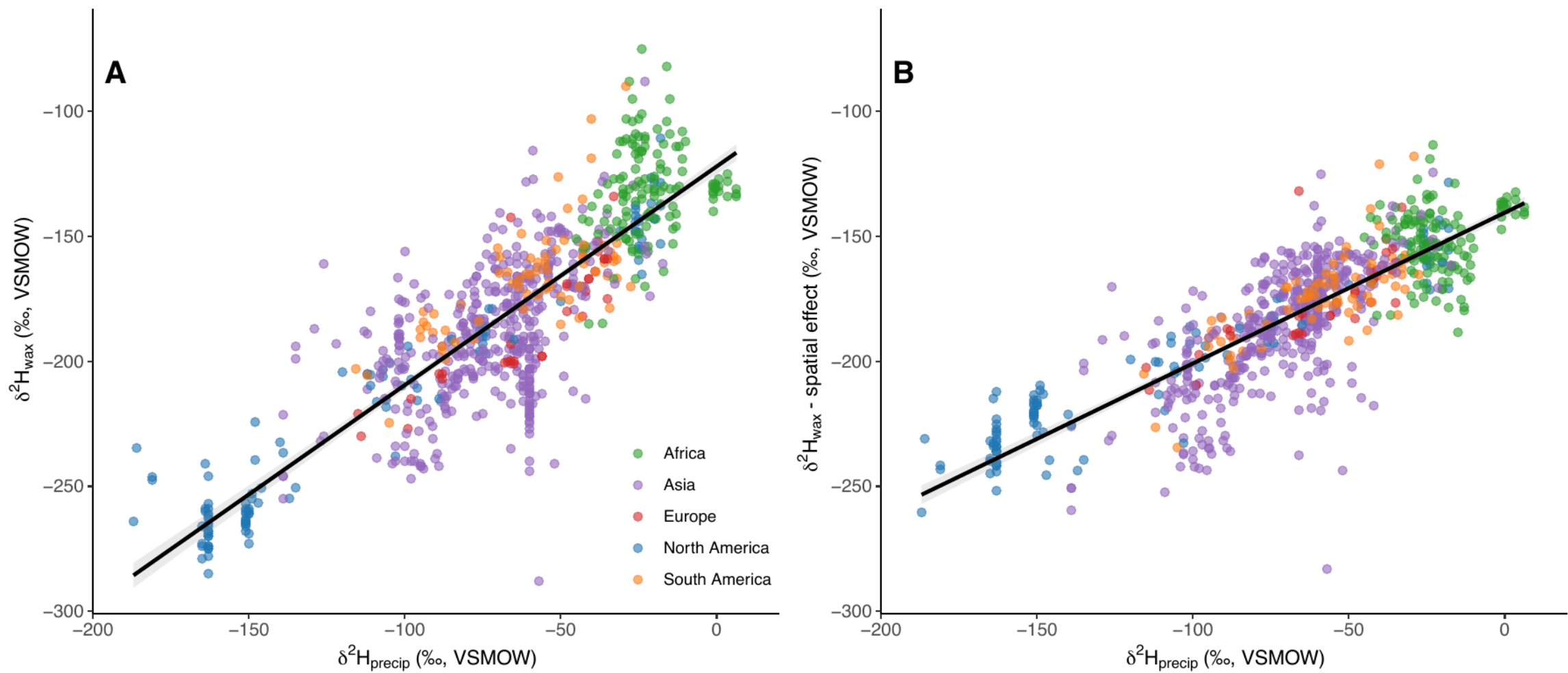
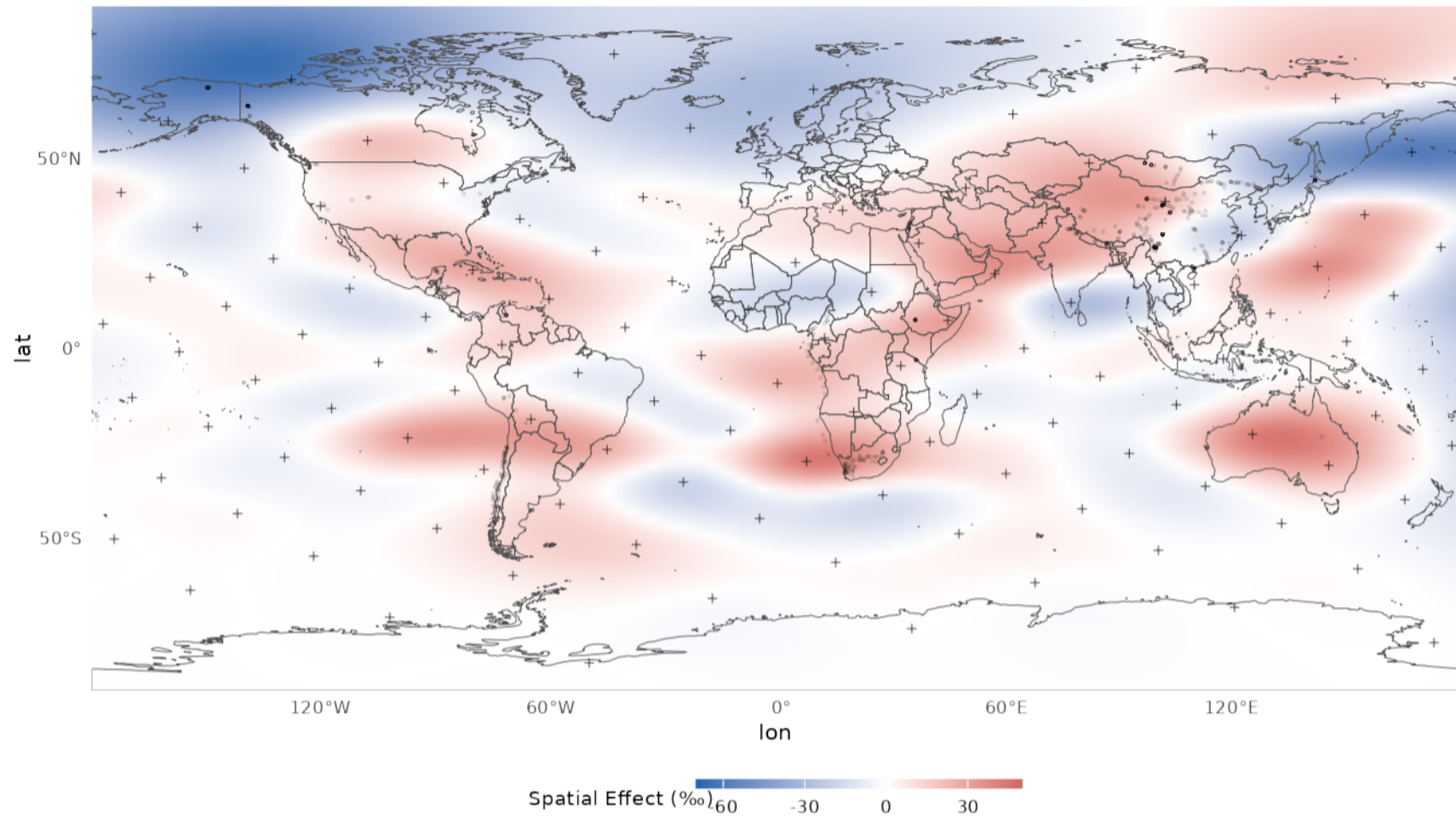


Figure 3: Spatial Patterns in Model Parameters (Full_sp Model)

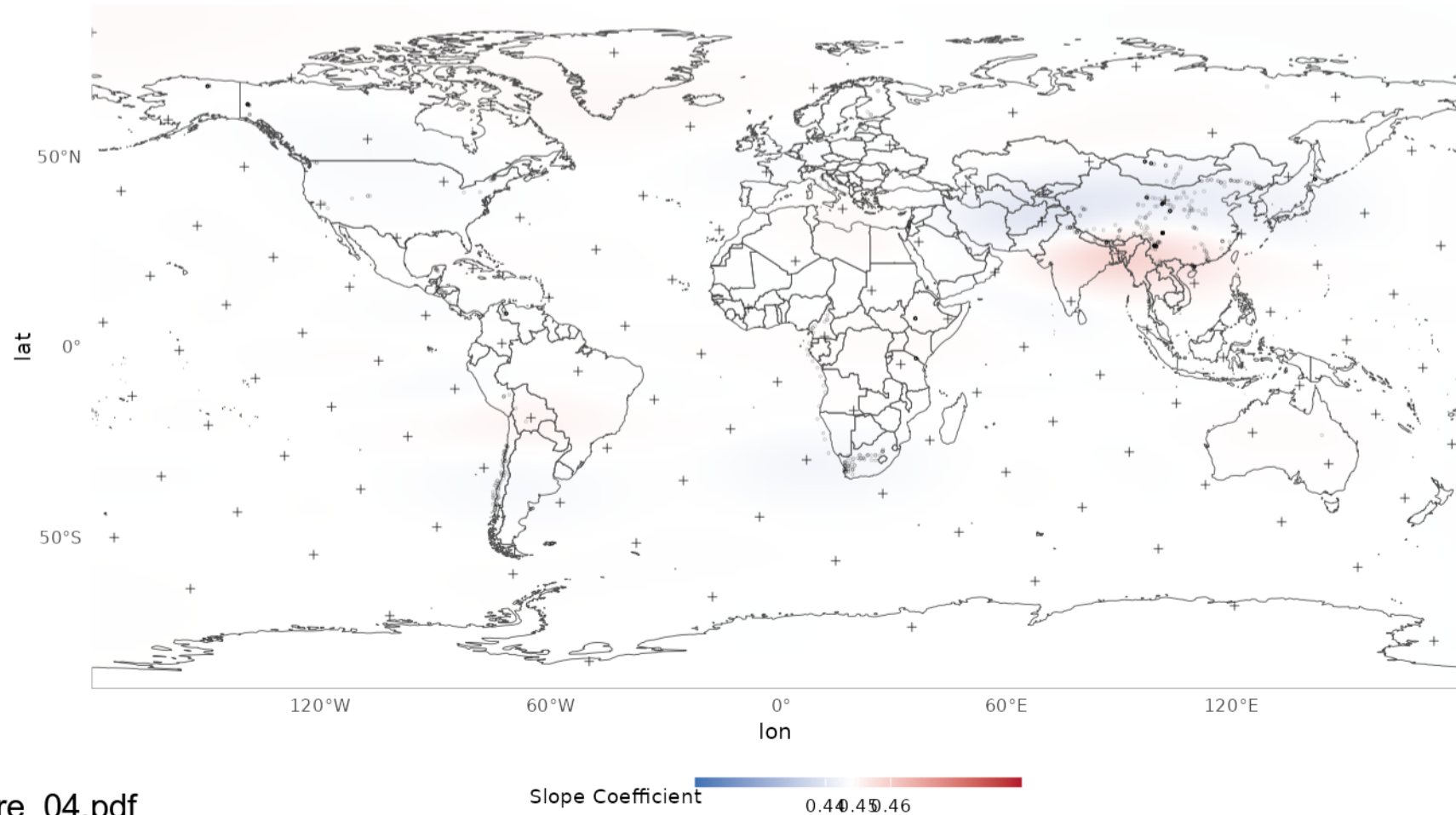
A. Spatial Intercept Variations

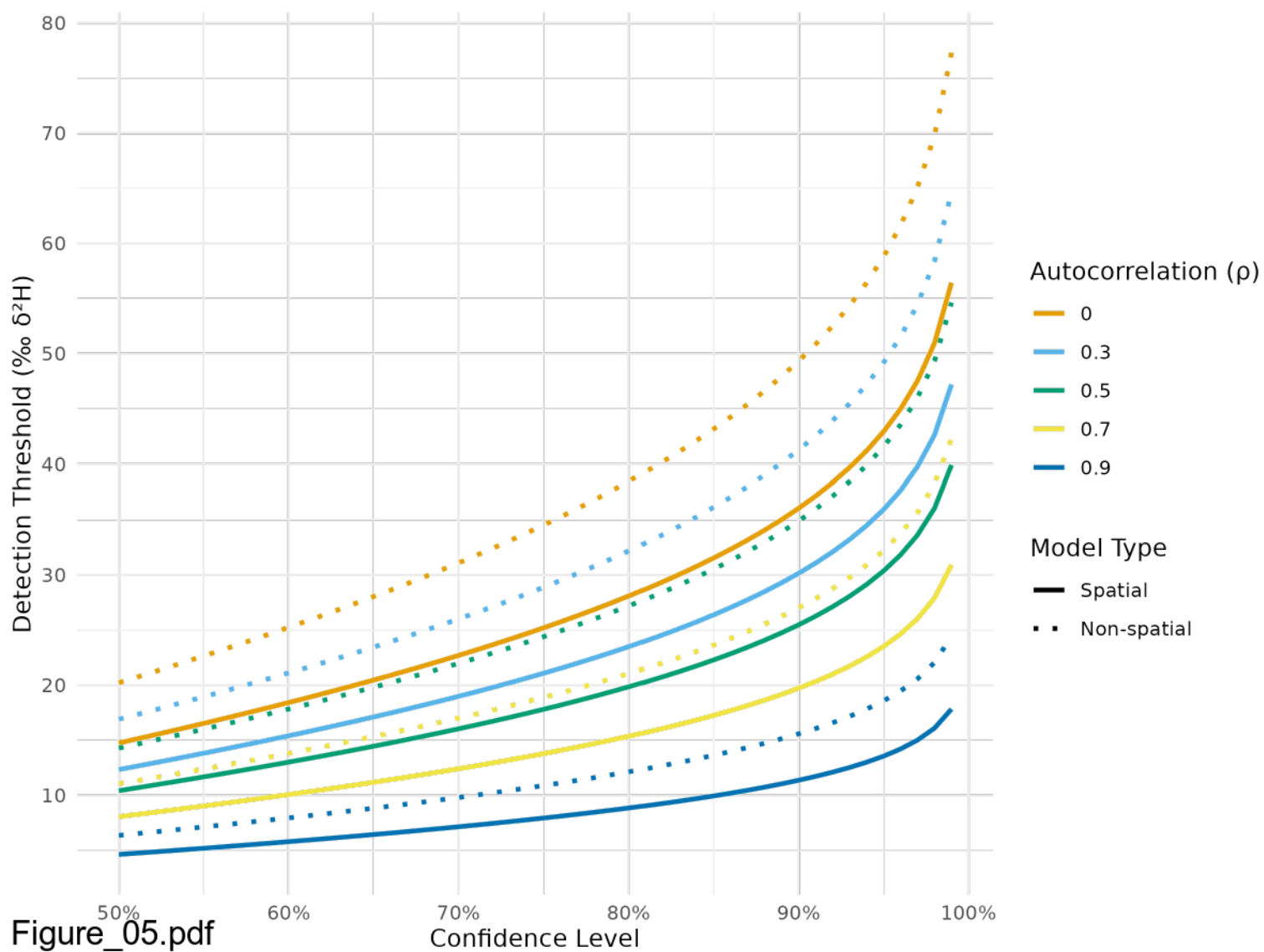
Range: -65.9 to 45.2 ‰ | SD: 16.3 ‰



B. Spatial Slope Variations

Range: 0.440 to 0.458 | Global: 0.448





Figure_05.pdf

Table 1: Model performance metrics for all candidate models

Model	Predictors	Max $\hat{R}$	Min ESS	LOOIC	SE	Δ LOOIC	SE	p_{eff}	$n_{\text{hi-k}}$
baseline	$\delta^2\text{H}_p$	1.004	3474	1336.7	53.0	493.5	54.8	3.6	0
baseline.sp	$\delta^2\text{H}_p$ + GP	1.008	1428	913.9	73.3	70.7	22.9	43.5	3
baseline.env	$\delta^2\text{H}_p$, elev, precip	1.004	1000	1235.7	58.1	392.5	44.7	14.0	0
baseline.env.sp	$\delta^2\text{H}_p$, elev, precip + GP	1.003	1428	860.6	81.5	17.4	7.5	52.8	2
baseline.veg	$\delta^2\text{H}_p$, PFT, C4	1.003	3473	1290.7	51.0	447.5	54.8	8.2	0
baseline.veg.sp	$\delta^2\text{H}_p$, PFT, C4 + GP	1.005	928	885.0	74.6	41.8	20.0	48.5	2
full	$\delta^2\text{H}_p$, PFT, C4, elev, precip	1.006	1129	1217.8	57.5	374.6	44.4	17.7	0
full.sp	$\delta^2\text{H}_p$, PFT, C4, elev, precip + GP	1.004	1581	843.2	80.4	0.0	0.5	55.9	2
full.interact	$\delta^2\text{H}_p$, PFT $\times$ $\delta^2\text{H}_p$, C4 $\times$ $\delta^2\text{H}_p$, elev, precip	1.010	971	1218.2	57.6	375.0	44.3	17.9	0
full.interact.sp	$\delta^2\text{H}_p$, PFT $\times$ $\delta^2\text{H}_p$, C4 $\times$ $\delta^2\text{H}_p$, elev, precip + GP	1.006	1743	843.2	80.3	0.0	0.0	56.0	2
elevation_only.sp	$\delta^2\text{H}_p$, elev + GP	1.005	1484	860.7	81.1	17.5	7.3	52.7	2
elevation_c4.sp	$\delta^2\text{H}_p$, elev, C4 + GP	1.003	1458	862.5	80.7	19.3	7.0	52.6	3
c4_only.sp	$\delta^2\text{H}_p$, C4 + GP	1.007	1703	916.1	72.4	72.9	23.9	41.7	2
elevation_c4.interact.sp	$\delta^2\text{H}_p$, C4 $\times$ $\delta^2\text{H}_p$, elev + GP	1.010	1440	862.7	80.9	19.5	7.1	52.4	2

Note: $\hat{R}$ = Gelman-Rubin convergence diagnostic; ESS = effective sample size; LOOIC = leave-one-out information criterion; SE = standard error; p_{eff} = effective number of parameters; $n_{\text{hi-k}}$ = number of high Pareto- k values; GP = Gaussian process; $\delta^2\text{H}_p = \delta^2\text{H}_{\text{precip}}$; elev = elevation.

Table 2: Global parameter estimates for all models

Model	RMSE		R^2		β_0		β_{OIPC}		λ	GP scale (km)	
baseline	21.1	[21.1, 21.2]	0.703	[0.701, 0.704]	-125.7	[-128.4, -122.9]	0.818	[0.783, 0.855]	8.3	[3.3, 18.2]	-
baseline_sp	15.9	[15.4, 16.3]	0.831	[0.822, 0.842]	-139.4	[-150.9, -128.1]	0.608	[0.511, 0.712]	7.2	[2.9, 15.5]	3937 [2453, 6051]
baseline_env	19.8	[19.6, 19.9]	0.740	[0.735, 0.744]	-125.6	[-128.5, -122.5]	0.833	[0.792, 0.875]	7.5	[3.5, 14.5]	-
baseline_env_sp	15.3	[15.0, 15.6]	0.844	[0.837, 0.850]	-153.4	[-167.6, -140.0]	0.494	[0.361, 0.616]	6.8	[3.0, 13.5]	3402 [2081, 5579]
baseline_veg	20.5	[20.4, 20.6]	0.720	[0.717, 0.721]	-120.1	[-123.9, -116.3]	0.808	[0.766, 0.853]	14.7	[5.2, 37.4]	-
baseline_veg_sp	15.6	[15.0, 16.1]	0.837	[0.827, 0.849]	-135.8	[-147.9, -123.6]	0.565	[0.457, 0.677]	8.3	[3.3, 18.0]	4247 [2680, 6128]
full	19.5	[19.4, 19.7]	0.746	[0.742, 0.750]	-120.5	[-125.0, -115.8]	0.827	[0.779, 0.877]	9.6	[4.3, 19.6]	-
full_sp	15.1	[14.8, 15.5]	0.848	[0.841, 0.854]	-151.4	[-167.5, -136.6]	0.444	[0.305, 0.576]	8.0	[3.4, 16.1]	3784 [2359, 5843]
full_interact	19.5	[19.4, 19.7]	0.746	[0.742, 0.750]	-120.5	[-125.1, -115.8]	0.827	[0.779, 0.875]	9.6	[4.0, 20.2]	-
full_interact_sp	15.1	[14.8, 15.5]	0.848	[0.841, 0.854]	-151.7	[-167.1, -136.9]	0.443	[0.304, 0.575]	7.9	[3.4, 16.1]	3785 [2382, 5873]
elevation_only_sp	15.3	[15.0, 15.6]	0.844	[0.837, 0.850]	-152.8	[-166.9, -139.3]	0.498	[0.365, 0.625]	6.8	[3.0, 13.5]	3412 [2083, 5540]
elevation_c4_sp	15.3	[15.0, 15.6]	0.844	[0.837, 0.849]	-152.2	[-166.6, -138.9]	0.500	[0.367, 0.625]	6.8	[3.0, 13.5]	3421 [2065, 5631]
c4_only_sp	16.0	[15.5, 16.4]	0.830	[0.822, 0.840]	-138.2	[-148.8, -127.7]	0.600	[0.508, 0.699]	7.0	[2.8, 14.8]	3824 [2340, 5939]
elevation_c4_interact_sp	15.3	[15.0, 15.6]	0.843	[0.837, 0.849]	-152.1	[-166.6, -138.9]	0.501	[0.372, 0.625]	6.8	[3.0, 13.3]	3429 [2077, 5651]

Note: Values shown as mean [95% credible interval]. RMSE = root mean squared error; β_0 = global intercept; β_{OIPC} = global slope with respect to $\delta^2\text{H}_{\text{precip}}$; λ = integration length scale; GP scale = Gaussian process length scale in kilometers.

Table 3: Variance decomposition for spatial models

Model	Spatial (%)	Residual (%)	Intercept (%)	Slope (%)
baseline_sp	54.4	45.6	97.1	2.9
baseline_env_sp	66.7	33.3	99.8	0.2
baseline_veg_sp	56.9	43.1	94.0	6.0
full_sp	68.2	31.8	99.7	0.3
full_interact_sp	68.3	31.7	99.7	0.3
elevation_only_sp	66.3	33.7	99.8	0.2
elevation_c4_sp	65.8	34.2	99.8	0.2
c4_only_sp	53.1	46.9	97.7	2.3
elevation_c4_interact_sp	65.7	34.3	99.8	0.2

Note: Variance components shown as percentage of total variance. Spatial = variance explained by spatial Gaussian process; Residual = unexplained variance; Intercept/Slope = proportion of spatial variance attributed to intercept vs. slope components.

Table 4: Environmental covariate coefficients

Model	β_{C4}	β_{tree}	β_{shrub}	β_{grass}	β_{precip}
baseline	-	-	-	-	-
baseline_sp	-	-	-	-	-
baseline_env	-	-	-	-	-0.053 [-0.078, -0.028]
baseline_env_sp	-	-	-	-	0.009 [-0.023, 0.042]
baseline_veg	-0.037 [-0.121, 0.047]	-0.005 [-0.007, -0.003]	0.001 [-0.001, 0.003]	-0.002 [-0.003, -0.001]	-
baseline_veg_sp	0.132 [0.030, 0.232]	-0.001 [-0.003, 0.001]	0.002 [-0.000, 0.004]	-0.002 [-0.004, -0.001]	-
full	-0.069 [-0.160, 0.022]	-0.003 [-0.005, -0.001]	0.001 [-0.001, 0.003]	-0.002 [-0.004, -0.001]	-0.027 [-0.055, 0.002]
full_sp	0.049 [-0.058, 0.157]	-0.002 [-0.004, 0.001]	0.002 [-0.001, 0.004]	-0.002 [-0.003, -0.001]	0.017 [-0.016, 0.050]
full_interact	-0.069 [-0.160, 0.024]	-0.003 [-0.005, -0.001]	0.001 [-0.001, 0.003]	-0.002 [-0.004, -0.001]	-0.027 [-0.055, 0.002]
full_interact_sp	0.049 [-0.057, 0.156]	-0.002 [-0.004, 0.001]	0.002 [-0.001, 0.004]	-0.002 [-0.003, -0.001]	0.017 [-0.017, 0.050]
elevation_only_sp	-	-	-	-	-
elevation_c4_sp	0.032 [-0.066, 0.129]	-	-	-	-
c4_only_sp	0.111 [0.014, 0.203]	-	-	-	-
elevation_c4_interact_sp	0.032 [-0.067, 0.131]	-	-	-	-

Note: Coefficient estimates shown as mean [95% credible interval]. β_{C4} = C4 grass fraction effect; β_{tree} , β_{shrub} , β_{grass} = plant functional type effects; β_{precip} = precipitation effect. Dashes indicate parameters not included in model.